DPP	DEPARTMENTAL POLICY AND PROCEDURE		
Title	**BLOOD TRANSFUSION THERAPY**		
Department	**Neonatology Department**		
Applies to	**All NICU Medical and Nursing Staff**		
Policy Code and Version Number	**DPP-NICU-003 (v2)**		
Last Version Approval Date	**28/1/2022**	Number of Pages	**7**

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APPROVAL DATE	EFFECTIVE DATE	TYPE OF REVISION	REVIEW DATE
19/ 3 / 2024	7/ 4 / 2024	TRIENNIAL	18 / 3 / 2027

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1.0 PURPOSE

1.1. To restore intravascular volume after hemorrhage.

1.2. To restore the oxygen-carrying capacity of blood by replacing red blood cells.

1.3. To replace clotting factor and correction of anemia.

2.0 DEFINITION/ABBREVIATION

2.1. **Blood transfusion therapy** is the intravenous administration of whole or blood component for therapeutic purposes.

3.0 RESPONSIBILITY

3.1. **NICU Physician**:

3.1.1. Write clear plan for procedure.

3.1.2. Gives the parents detailed explanation about the procedure.

3.1.3. Observe patient pre, during and post procedure.

3.2. **NICU Nurse**:

3.2.1. Confirms the physician's order, signed consent, and patient identification accurately.

3.2.2. Coordinate with the blood bank for laboratory requests and cross matching.

3.2.3. Monitors patient's vital signs.

3.2.4. Safe handling and storage of received blood.

3.2.5. Safe administration of blood.

3.2.6. Monitors patient's vital signs during procedure.

3.2.7. Reports any untoward events during the procedure to the physician.

3.2.8. Cares for the patient during the course of the whole procedure.

4.0 POLICY

4.1. Blood transfusion must have a written physician's orders for type, amount, and rate of blood administration.

4.2. Consent for blood transfusion must be signed by parents for children.

4.3. Standard precautions are to be followed when handling blood product.

4.4. Blood for transfusion must be cross matched to the recipient's blood type, and two qualified staff nurses should verify the patient's identity prior to blood extraction for type and cross match.

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4.5. Blood must be counterchecked by the Doctor and the attending Nurse, once Blood is obtained from the blood bank.

4.6. Before giving blood transfusion two nurses should identify the patient correctly using the appropriate means of identification such as using medical record number, identification bracelet, and patient's name.

4.7. Blood transfusion must be checked at patient's bedside by two registered nurses or a registered nurse and a physician before infusion.

4.7.1. Name of patient

4.7.2. Chart number

4.7.3. Serial Number on the Blood bag level

4.7.4. Expiration date

4.7.5. Blood Group compatibility

4.7.6. Serology – free results or negative results

4.8. Baseline vital signs should be taken and recorded.

4.9. Standard blood filter must be used and to be changed after 8 hours.

4.10. Nurse should observe patient closely for the first 5 to 15 minutes after the blood transfusion is initiated.

4.11. The blood transfusion is to begin slowly within 30 minutes after obtaining the blood from the blood bank and should be infused within 2 to 3 hours but not more than 4 hours.

4.12. Blood transfusion should be stopped and physician should be notified immediately if signs of blood transfusion reaction occur.

4.13. Medication should never be injected into an Intravenous line with the blood component because of the risk of contaminating the blood product with bacteria.

5.0 PROCEDURE (S)

5.1. Verify a written doctor's order for Blood Transfusion.

5.2. Assess the patient for the following:

5.2.1. History of blood reaction

5.2.2. Check blood return for venous access.

5.2.3. Obtain vital signs and document

5.3. Check that transfusion consent has been properly signed and explain to the patient.

5.4. Explain procedure to patient and family if present, particularly the need for frequent vital sign checks

5.5. Pre medicate the patient as per doctor' order, 15 to 30 minutes before transfusion. (Optional)

5.6. Obtain blood product by registered nurse from blood bank & check the following with a doctor:

5.6.1. Serial Number of blood

5.6.2. Expiration date

5.6.3. Blood Group

5.6.4. Negative Serology Reports

5.6.5. Check blood product for clots & haemolysis.

5.7. Confirm the identity of the patient and the blood product by two staff nurses, or a staff nurse and a physician before initiating the blood

5.7.1. Name of patient.

5.7.2. Chart number.

5.7.3. Serial Number of blood

5.7.4. Expiration date

5.7.5. Blood Group compatibility. All information must match the medical record and the patient.

5.7.6. Negative Serology report

5.7.7. Check blood product for clots & haemolysis

5.8. Wash hands and wear disposable gloves.

5.9. Prime the blood administration set with Normal Saline, ensure that it flush through the IV tubing to clear air bubbles (optional for adult)

5.10. Invert blood component bag gently 2 to 3 times

5.11. Spike the product bag using aseptic technique with clamp close. Squeeze drip chamber, allowing saline to cover the filter.

5.12. Close the clamp on the Normal Saline and open the clamp on the blood. Allow the blood to completely cover the filter.

5.13. Remain with patient during the first 5 to 15minutes after initiating the blood transfusion

5.14. If no signs of reaction for the first 15 minutes, regulate the flow rate according Physician's order to run at least 2 to 4 hours (drop factor of BT is 10 drops/ml).

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5.15. Monitor vital signs according to blood transfusion observation sheet.

5.16. Observe for chills, flushing, dyspnea, rash or other signs of transfusion reaction

5.17. Stop blood transfusion immediately for any adverse hemolytic reaction.

5.17.1. Assess the patient

5.17.2. Prioritize your nursing intervention according to your assessment.

5.17.3. Inform the doctor

5.17.4. Inform the CN/HN/Supervisor

5.17.5. Write an Incident report

5.18. Flush the IV tubing with Normal Saline and discard the blood bag according to bio-hazardous waste disposal, when transfusion is completed

5.19. Document in the nurse's notes:

5.19.1. Date & time blood started & consume

5.19.2. Amount & Type of blood infused

5.19.3. Initial & subsequent vital signs & the response of the patient.

5.19.4. IV cannula size, location & condition of Intravenous (IV) site.

5.19.5. Instructions given & patient understanding of instructions.

5.20. **BLOOD TRANSFUSION:** The transfusion of blood and blood products remains a highly effective and potentially lifesaving treatment for many patients. However, blood transfusion rom one individual to another is risky; significant among these risks is the potential for human error and subsequent transfusion of the incorrect blood component.

5.21. Blood Samples to be obtained for Acute Blood Reaction:

5.21.1. Blood sample to examine serum for hemoglobin and confirm RBC group and type.

5.21.2. Anticoagulated blood sample for a direct Combs' test to determine the presence of antibody on the RBCs.

5.21.3. The first voided urine sample to test for hemoglobinuria

5.21.4. Blood Components:

5.21.4.1. Platelet – consist of platelets suspended in Plasma. Infused 20-60 mins; depending on total volume, 1 unit of platelet / 10 kgs. of body weight. Indications include prevention or resolution of hemorrhage in patients with thrombocytopenia or platelet dysfunction.

5.21.4.2. Plasma (Fresh or Fresh Frozen)

5.21.4.3. Consist of water (91%), plasma proteins including essential clotting factors (7%), and carbohydrates (2%).

5.21.4.4. Infusion can be completed within 15-30 mins., depending on the total volume

5.21.4.5. Indications include treatment of blood loss or blood clotting disorders related to liver disease and failure. DIC (Disseminated Intravascular Coagulation), over anticoagulation with warfarin, congenital or acquired clotting factor deficiencies

5.21.4.6. Cryoprecipitate – consist of certain clotting factors suspended in 10-20 ml plasma. Each unit contains 80-120 units of factor VIII (antihemophilic and von Willebrand factors), 250 mg fibrinogen, and 20% -30% of the factor XIII present in a unit of whole blood.

5.22. Indications include correction of deficiencies of factor VIII (ex. Hemophilia A and von Willebrand disease), factor VIII, and fibrinogen (ex. DIC).

5.23. Dosage: Adult dosage is 10 units, which may be repeated every 8-12 hours until the deficiency is corrected.

6.0 EQUIPMENTS

6.1. Gloves.
6.2. Syringes.
6.3. IV tube.
6.4. Cannula.
6.5. Blood transfusion Set.
6.6. Normal saline.

7.0 ATTACHMENT/FORMS

7.1. Consent for Blood / Blood Components Transfusion (GDOH-COR-CBCT-357)
7.2. Nurse Observation Form (Form No. 43)
7.3. Nurses Progress Notes (GDOH-NUR-NPN-203)
7.4. Blood and Blood Products Request and Release Form (GDOH-LAB-BBPR-319)
7.5. Physician Order Form (GDOH-INP-POF-079)

8.0 REFERENCES

8.1. MOH-KSA Guidelines for Neonatal Care

8.2. Clinical Guidelines for Neonatal care.

8.3. CBAHI Standards NICU 3rd version

8.4. Lippincott Manual of Nursing Practice, 7th edition by Nettina

8.5. Nurses' Guide to Clinical Practice, 5th edition by Temple & Johnson

9.0 REVISION CLAUSE

9.1. Modification of policy format, lay out and policy coding

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وزارة الصحة
Ministry of Health

Hospital: _____ :مستشفى

Region: _____ :المنطقة/المحافظة

Dept./Unit: _____ :القسم/الوحدة

MRN: | | | | | | | | | | | | :رقم الملف الطبي

Name: _____ :الاسم

Nationality: _____ :الجنسية

Age: _____ Years سنة ____ Months شهر ____ Days يوم :العمر

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ :تاريخ الميلاد

Gender: ☐ Male ☐ Female :الجنس

CONSENT FOR BLOOD/BLOOD COMPONENTS TRANSFUSION موافقة على نقل الدم/ مشتقات الدم

I, the undersigned, _____ in my Capacity as

☐ **Patient** ☐ **Legal Guardian** ☐ **Relative**

(Specify: _____) of

(Patient's Name: _____)

That I have ..... I need the

Following units of .....

Since this will support the treatment of my condition.

It has been explained to me about the:

- Nature of the procedure.
- Benefits of this procedure.
- Risks involved with this procedure.
- Potential risks of not carrying out this procedure.
- Possible alternative treatment modalities.

I knowledge that all relevant information about the procedure has been given to me in the language that I understand,

I have the opportunity to ask questions concerning me / my patient's condition and about the procedure and all questions have been answered to my satisfaction.

Patient's/Guardian's Name: .....

Relationship to patient: .....

Signature;

Date:...../...../.....

I have seen this consent before starting blood/ blood components transfusion and explained this to patient/ guardian.

Physician's Name: .....

Signature:

Stamp:

Witness's Name: .....

Signature:

Date:...../...../.....

أنا الموقع أدناه: _____ بصفتي

☐ **المريض/** ☐ **ولي الأمر/** ☐ **القريب**

صلة القرابة:(_____)

اسم المريض:(_____)

اعاني من .....

وحاجتي إلى الوحدات التالية من .....

وذلك حسب حالتي المرضية كون هذا الدم سيساعد في التغلب على مرضي.

لقد تم الشرح التام لي عن مايلي:

- طبيعة الاجراء .
- الفوائد المرجوة من هذا الاجراء .
- المخاطر المرتبطة بهذا الاجراء .
- المخاطر المترتبة على عدم عمل هذا الاجراء .
- البدائل الممكنة للعلاج .

وأقر بأن كافة المعلومات الهامة ذات الصلة بهذا الإجراء قد تم ايضاحها لي بلغة مفهومة من قبلي، كما اتيحت لي الفرصة لطرح الاسئلة التي تخطر لي / لمريضي، وأنه قد تمت الإجابة على جميع استفساراتي بصورة مرضية.

إسم المريض/ولي الأمر : .....

صلة القرابة: .....

التوقيع:

التاريخ:...../...../.....

اطلعت على هذا الإقرار قبل إجراء نقل الدم ومشتقاته وقمت بشرح ذلك للمريض/ولي أمره.

اسم الطبيب: .....

التوقيع :

الختم :

اسم الشاهد: .....

التوقيع:

التاريخ:...../...../.....

Ministry of Health
General Directorate of Health Affairs
Al-Baha

وزارة الصحة
المديرية العامة للشئون الصحية بمنطقة الباحة

HOSPITAL :

Health Care No .: ☐☐☐☐☐☐☐☐☐☐☐☐

Name : .....

Age : ☐☐ Yrs./Month Sex : ☐M ☐F

Nationality : .....

Consultant In-Charge : .....

Dept : ..... Unit : .....

NURSES OBSERVATION FORM نمـوذج مـلاحظـات التمـريضـ

Nurse / Date	Time	Temp.	Puls Rate	Resp Rate	B.P.	Pupils		Level of Consciousness Muscle Tone &Reflex	Cough Reflex	Response to Stimuli		Perception Pulses			Skin Color	Rema-rks	Sign.
						R	L			Verbal	Painful	ORAL	Audio Aural	visual			

SPECIFIC NURSING CARE COMMENTS :

.....

.....

.....

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MRN: | | | | | | | | | | رقم الملف الطبي:

Name: _____ الاسم:

Nationality: _____ الجنسية:

Age: _____ ☐ سنة Years ☐ شهر Months ☐ يوم Days العمر:

Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

NURSES PROGRESS NOTES FORM

DATE &TIME	NURSES NOTES (D-data A-action R-response)	REGISTERED NURSE INITIAL WITH ID NUMBER

Note: Write the time in each entry & affix your initial at the end of each paragraph. Document your complete Name, Initial, Job number, Date & Time at the closure of your documentation. Draw a line across empty spaces.

0 000000 002035

SN

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Date of Birth: ____/____/14____ H ____/____/20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

BLOOD & BLOOD PRODUCTS REQUEST & RELEASE FORM

Note: One form must be submitted for each unit ordered. Blood release copy remains on chart until blood is needed. Blood bank copy must be returned to blood bank on completion of transfusion or return of blood.
RETURN BLOOD TO CENTRAL BLOOD BANK WITH 30 MINUTES FROM RELEASE OF BLOOD FROM BLOOD BANK IF NOT TRANFUSED.

Requisition for: ☐ Packed Red Cells ☐ Whole Blood ☐ Fresh Frozen Plasma ☐ Cryoprecipitate ☐ Platelet Concentrate

Special Instructions: ☐ Irradiated ☐ Leukocyte poor (Filtered) ☐ Volume _____ml

Requesting Physician's name: _____ Stamp&Signature: _____ Date:____/____/____

RELEASE BLOOD PRODUCTS

Unit No _____ Product _____ Blood Group _____ Expiry Date:____/____/____

Blood unit issued by: Name _____ Sign _____ Date:____/____/____ Time _____

Blood unit received by: from blood bank: Name: _____ Signature: _____

Blood unit received in the ward by: Name: _____ Signature: _____ Time: _____

TRANSFUSION RECORD

(This part must be completed when transfusion is started)

Pre transfusion

Patient's temp. _____ BP _____ Pulse _____

Date of transfusion _____ Time _____

I have verified ID, Blood Group and Rh of the recipient and donor on bag label and blood issue and compatibility testing forms. I have further verified that the recipient is same person named on all documents.

Remarks: _____

_____
Name and signature of person starting transfusion

_____
Name and signature of 2nd verifying person

(This part must be completed if blood is returned to BB)

Reason for return _____

Date ____/____/____ Time _____

Returned by: Name and signature _____

Received in blood bank :Date ____/____/____ Time ___

By Name and Sign _____

(I verify temperature and integrity of bag is acceptable)
If unacceptable for reuse, fill incident report form.

(This part must be completed when transfusion is stopped)

Post transfusion

Ending time: _____ Amount given _____

Patient's temp. _____ BP _____ Pulse _____

_____
Name and signature of person completing transfusion

Transfusion Reaction: ☐ None

☐ Yes (Specify)_____

If transfusion reaction is suspected then immediately:

1. Discontinue transfusion.
2. Treat shock, if present.
3. Notify attending physician.
4. Notify blood bank at once and send to blood bank/ lab.
   a. Unused blood, transfusion set with tubing attached (remove needle)
   b. Recipient venous blood sample 5ml EDTA tube.
   c. Completed transfusion reaction form.
   d. Next voided urine to central lab.

Other difficulties: (Equip, clots etc) ☐ None

☐ Yes (Specify)_____

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Date of Birth: ____ / ____ / 14____ H ____ / ____ / 20____ تاريخ الميلاد:

Gender: ☐ Male ☐ Female الجنس:

PHYSICIAN ORDER FORM

ALLERGIC TO:

DATE	TIME	PHYSICIAN ORDERS	PHYSICIAN STAMP / SIGNATURE	RN INITIAL / JOB NUMBER